THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL-I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

December 10, 2013 Time: 2:30 - 5:30pm

I. Answer ALL questions. Section A carries 60 marks, and Section B carries 40 marks.

II. Atomic weights: H 1; C 12; F 19; O 16; Na 23; Mg 24.3; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (20 multiple choice questions with 3 marks each)

1. Which of the following statements is more appropriate? Compared to groundwater, surface water often contains (a) more suspended solids, (b) greater hardness, (c) more volatile organic compounds, or (d) less pathogen.
2. Which of the following treatment processes is most appropriate for water softening? (a) Primary sedimentation, (b) secondary sedimentation, (c) lime-soda treatment, (d) aeration, or (e) sand filtration.
3. A eutrophic lake has an N/P ratio (by weight) of 3:1. Growth of algae in the lake is more likely to be limited by (a) N, (b) P, (c) C, (d) O, or (e) cannot tell due to insufficient information.
4. Which of the following is most suitable for secondary disinfection? (a) Ozone, (b) UV light, (c) chlorine dioxide, or (d) chloramine.
5. An ultrafiltration (UF) membrane has a 4-log removal of virus. The membrane has a virus rejection of (a) 99%, (b) 99.9%, (c) 99.99%, (d) 99.999%, or (e) cannot tell due to insufficient information.
6. You are asked to provide a back-of-the-envelope estimate of the Henry's constant ($K_{H,O2}$) for dissolved oxygen in a typical surface water. Which is the closest value of $K_{H,O2}$? (a) 1 mg/L·bar, (b) 10 mg/L·bar, (c) 50 mg/L·bar, (d) 200 mg/L·bar, or (e) 1000 mg/L·bar.
7. Which of the following methods is least effective in enhancing the removal efficient of a rectangular sedimentation tank? (a) Increasing the length of the tank, (b) increasing the width of the tank, (c) increasing the depth of the tank, or (d) reducing the flow rate of the incoming water.
8. A water contains 20 mg/L Na^+, 12 mg/L Mg^{2+}, 30 mg/L Ca^{2+}, 56 mg/L Cl^-, and 110 mg/L HCO_3^-. Its pH value is around 7.8. What is the carbonate hardness of the water (expressed by $CaCO_3$ equivalent)? (a) 42 mg/L, (b) 54 mg/L, (c) 62 mg/L, (d) 90 mg/L, (e) 110 mg/L, or (f) 125 mg/L.
9. A wastewater has a BOD_3 of 120 mg/L and BOD_5 of 150 mg/L. Which is the closest value of the ultimate BOD of the wastewater. (a) 30 mg/L, (b) 135 mg/L, (c) 150 mg/L, (d) 175 mg/L, or (e) 270 mg/L.
10. The Streeter–Phelps oxygen sag curve is often used to determine the impact of wastewater discharge on the dissolved oxygen level in a river downstream to the discharge point. According to the Streeter–Phelps model, oxygen depletion is more likely to occur (a) for faster moving rivers, (b) for shallower rivers, (c) when the BOD concentration in the wastewater discharge is reduced, (d) if the organic matter in the wastewater is less biodegradable, or (e) in warmer days.
11. In natural marine waters, organic nitrogen (Org-N) from sewage disposal would normally go through the transformation of (a) Org-N → Ammonia-N → Nitrite → Nitrate, (b) Org-N → Ammonia-N → Nitrite → N_2, (c) Org-N → Nitrite → Ammonia-N → Nitrate, or (d) Org-N → Ammonia-N → N_2 → Nitrate.
12. Which of the following description of asbestos is false: (a) an indoor air pollutant, (b) a product of incomplete combustion, (c) a natural product, or (d) an inorganic compound.
13. Which of the following description of dioxins is false: (a) containing two oxygen atoms, (b) incineration byproducts, (c) chlorinated compounds, or (d) emitted by cigarette smoking.
14. Chlorofluorocarbons (CFCs) are known to cause the depletion of stratospheric ozone. They had been widely used in numerous industrial applications due to certain physicochemical properties until the recent decade. Which of the following descriptions

of CFCs is false: (a) not harmful to health, (b) water soluble, (c) not flammable, or (d) not reactive.

15. Adiabatic lapse rate is a measurement of the atmospheric temperature drop for the increase of altitude. Which of the following statement is false? According to thermodynamics, adiabatic lapse rate is dependent on (a) humidity, (b) specific heat of air, (c) altitude, or (d) carbon dioxide content.
16. Degradation rate of organic pollutant is a function of pollutant concentration. This is most often expressed by the two parameter Monad equation. Which of the following statement is inaccurate, according to Monad equation: (a) degradation rate drops as time goes to infinitive; (b) degradation rate increases linearly at low substrate concentration; (c) one of the parameter is the substrate concentration at which the rate is half of the maximum degradation rate; (d) degradation rate approaches a maximum value as substrate concentration continues to increase.
17. A concentration level for NO_2 is at 300 $\mu g/m^3$. Given that one mole of an ideal gas occupies 24.5 litres at 25°C and 1 atm, and the molecular weights of N and O are 14 and 16, respectively. Express the concentration level of NO_2 in ppm: (a) 0.16 ppm, (b) 0.15 ppm, (c) 0.18 ppm, or (d) 0.24 ppm.
18. The most commonly used systems for treatment of exhaust gases from motor vehicles are thermal reactors, exhaust gas recirculation (EGR) systems, and catalytic converters. Which one of the following statements is incorrect for EGR systems: (a) it lowers the combustion temperature, (b) it may reduce the production of NO_X, (c) it may affect the air-to-fuel ratio, or (d) it circulates portion of the exhausted gas back into the incoming air/fuel mixture.
19. Traffic noise at a location due to a nearby highway has a present value of 65.5 dB(A). In two years time, the traffic volume in the highway will increase by 35%. What will be the traffic noise level at the location? (a) 88.4 dB, (b) 76.1 dB, (c) 68.1 dB, or (d) 66.8 dB.
20. Electrostatic precipitators use very high negative voltage normally in the range of (a) 1000-10,000 V, (b) 10,000-100,000 V, (c) 100,000-1,000,000 V or (d) higher than 1,000,000 V.

Section B (Two questions with 20 marks each)

21. Disinfection is a critical step for drinking water treatment. Chlorine (in the form of chlorine gas or hypochlorite salts) and ozone are commonly used disinfectants.

(a) A potential concern of chlorination is the formation of disinfection by-products (DBPs) that pose significant risk to human health. Briefly discuss the issue of DBPs. Recommend rational approaches for controlling DBPs.

(6 Marks)

(b) *Cryptosporidium* is a pathogen that can cause Cryptosporidiosis – a gastrointestinal illness with severe diarrhea symptoms. In 1993, an outbreak of Cryptosporidiosis in Milwaukee in the United States caused more than 100 deaths, and nearly a half million illness cases were reported. *Cryptosporidium* can be inactivated by applying ozone or chlorine dioxide.

(i) Assume that a disinfection tank can be modeled as a continuous stirred tank reactor where contaminant concentration is identical everywhere inside the reactor. Show that:

$$N = N_0/(1+kCT)$$

where N is the active pathogen (i.e., *Cryptosporidium*) concentration in the effluent, N_0 is that in the influent, k is the coefficient of specific lethality, C is the concentration of disinfectant, and T is the hydraulic retention time.

(ii) When ozone is applied for disinfection at 25°C, a CT value of 2.5 mg.min/L is required to achieve 90% inactivation of *Cryptosporidium*. Determine the coefficient of specific lethality for ozone against *Cryptosporidium*. If the disinfection tank has a volume of 200 m^3 and treats 10,000 m^3 of water per day, what is the required ozone concentration to achieve a 99.9% inactivation of *Cryptosporidium*?

(iii) Compared to ozone, chlorine dioxide requires greater CT value for disinfecting *Cryptosporidium* (e.g., 75 mg.min/L for 90% inactivation at 25°C). Which disinfectant (ozone or chlorine dioxide) is more effective against *Cryptosporidium*?

(14 Marks)

22. The public has been alerted in recent years by the increasing evidences of global climate change. There are two major factors affecting the global climate change - astrological factor and anthropogenic factor.

(a) Describe and discuss the three primary earth's orbital cycles which have profound impacts on the global climate change.

(10 marks)

(b) Name and discuss three most significant greenhouse gases (excluding water vapor) produced due to human activities, and their effects on the global temperature increase in recent decades.

(10 marks)

- END OF PAPER -